

# Communication lower bounds for Multi-Party Graph Traversal

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## Abstract

A large number of computational problems can be modelled as communication games: For example, given a graph, two players each have a subgraph they know is traversable, and wish to collaboratively construct a path from one vertex to another. Communication complexity studies such situations by creating asymptotic bounds on the minimum communication necessary to complete these tasks. In this report, we prove communication lower and upper bounds for a variety of graph traversal problems, such as the example above. We also review potential applications of these bounds, such as for modelling real-world scenarios and designing algorithms.

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## Motivation

We motivate our problem by considering a “Learning Graph”, and multiple agents which may wish to identify characteristics of the graph.

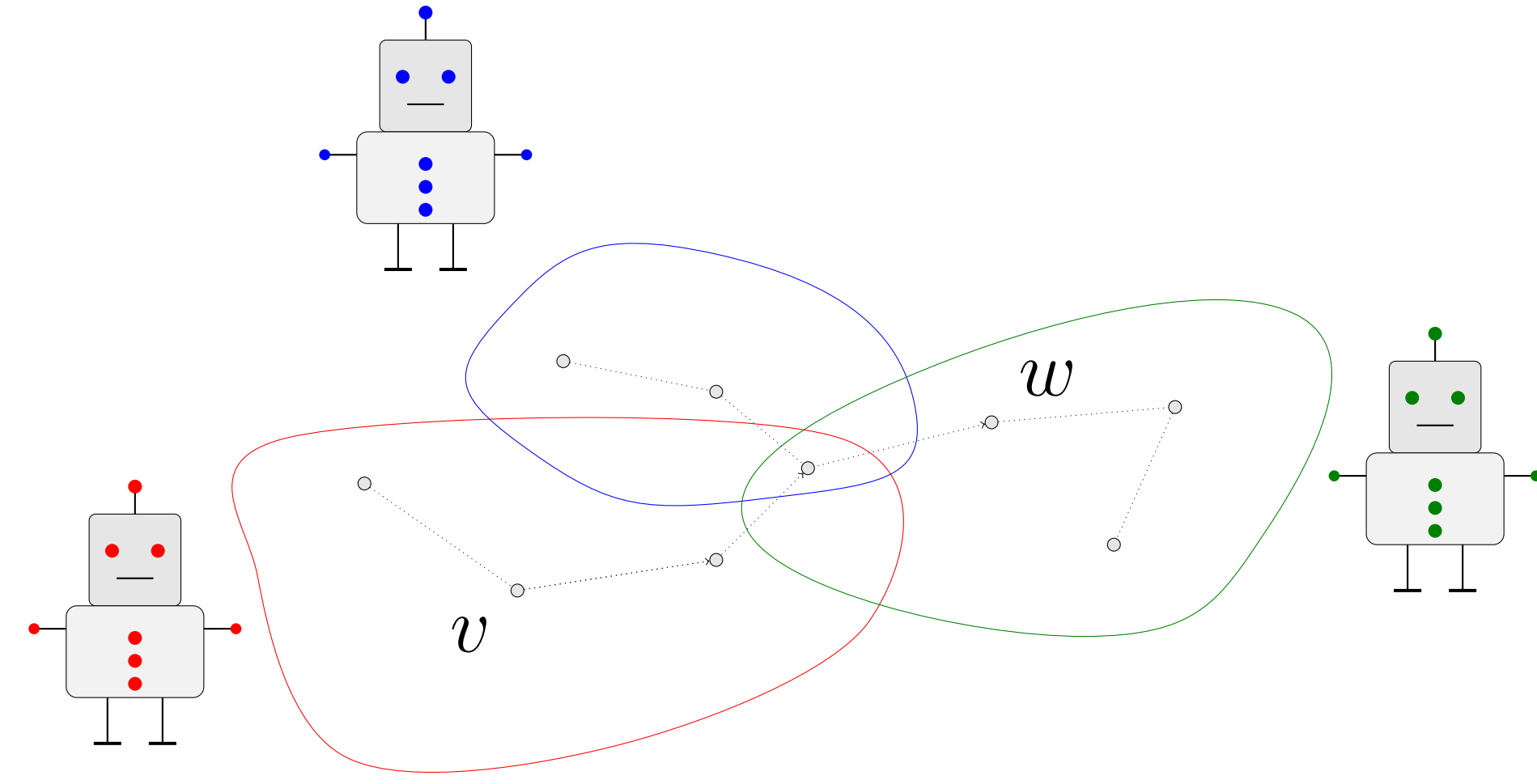


Figure 1: Multiple learning agents with graph ownership

## Basic Example: Equality

There are two players, traditionally called **Alice** and **Bob**.

- Alice holds input  $x$
- Bob holds input  $y$
- Together, they want to compute a function  $f(x, y)$

They are allowed to send messages (bits) back and forth. The goal is to minimize the total number of bits communicated.

**Equality** wishes to compute if two bit-strings are the same:

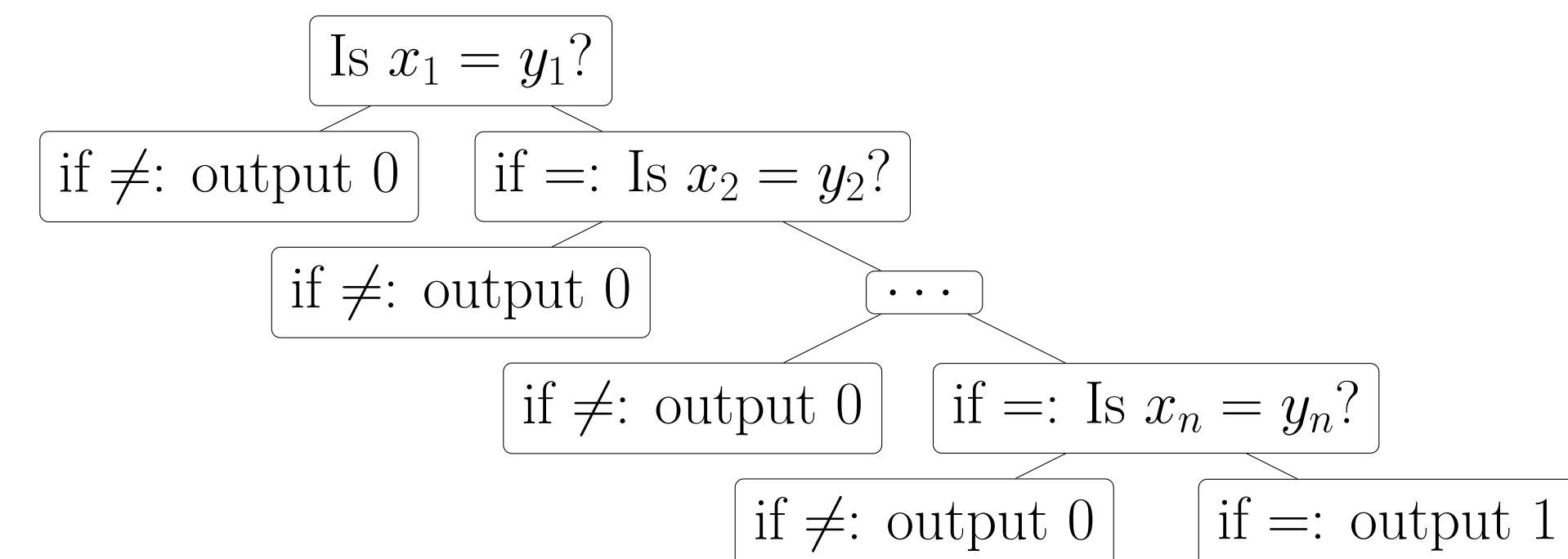


Figure 2: Protocol tree Equality

Figure 2 demonstrates how, in the worst case, we must check every bit, since if every bit except  $x_n$  and  $y_n$  are the same, we will have to check all  $n$  bits to show equality. Thus, the deterministic communication complexity of Equality is  $D(\text{EQ}) = n$

## Two Player TRAV with EQ

The following materials were required to complete the research:

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem
- Eu facilisis est tempus quis

The materials were prepared according to the steps outlined below:

- 1 Curabitur pellentesque dignissim
- 2 Eu facilisis est tempus quis
- 3 Duis porta consequat lorem
- 4 Curabitur pellentesque dignissim

## Multi-Player TRAV, disjoint graphs

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$E = mc^2 \quad (1)$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\cos^3 \theta = \frac{1}{4} \cos \theta + \frac{3}{4} \cos 3\theta \quad (2)$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\kappa = \frac{\xi}{E_{\max}} \quad (3)$$

## Two Player TRAV with DISJ.

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## Multi-Player TRAV, connected graphs



Figure 3: Figure caption

Nunc tempus venenatis facilisis. Curabitur suscipit consequat eros non porttitor. Sed a massa dolor, id ornare enim:

| Treatments  | Response 1 | Response 2 |
|-------------|------------|------------|
| Treatment 1 | 0.0003262  | 0.562      |
| Treatment 2 | 0.0015681  | 0.910      |
| Treatment 3 | 0.0009271  | 0.296      |

Table 1: Table caption

## Two-player TRAV, iterative

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## Multi-Player TRAV, iterative

Maecenas ultricies feugiat velit non mattis. Fusce tempus arcu id ligula varius dictum.

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem

## References

## Acknowledgements

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